NM Laboratory Activity 02 — Results

Fitting a Curve to the Dam | Student: Samson | 96 readings | Sampling interval: 0.25 h

1. Selected Model

Four-parameter logistic
Equation: h(t) = c + a / (1 + exp(-k(t - t■)))
The model was selected because the reservoir record shows a single smooth filling transition with a sigmoidal shape and one main inflection point. The choice was based on the observed physical/shape behavior, not on R^2 alone.

2. Fitted Parameters and Statistical Results

Parameter	Estimate	Unit	Standard error	t statistic	Two-tailed p-value	Conclusion
c	14.10600623	m	0.05016887	281.17048	< 1×10■¹■	Significant
a	0.48529909	m	0.08958411	5.41725	4.8×10■■	Significant
k	0.11998802	1/h	0.02650416	4.52714	1.786×10■■	Significant
t■	10.99008254	h	0.89853351	12.23113	< 1×10■¹■	Significant

Statistic	Value	Statistic	Value
SSE	0.01859995	SST	0.82449063
R^2	0.97744068	Standard error of estimate	0.01421877 m
n	96	p	4
Degrees of freedom (n – p)	92	Sampling interval	0.25 h

3. Derivative Results

Maximum finite-difference dh/dt: 0.120000 m/h.
Maximum continuous fitted dh/dt: 0.01455752 m/h at **10.99008 h** elapsed hours.
For the logistic model, the maximum continuous filling rate occurs at the inflection point t■.

Second-derivative interpretation: The second derivative indicates that the inflow/filling rate increases before the inflection point, is approximately zero at the inflection point, and decreases afterward as the reservoir approaches its higher settling level.

4. Residual Analysis

The residuals are centered around zero overall, but the residual sequence contains runs of seven consecutive positive residuals and seven consecutive negative residuals, indicating some systematic departure from the fitted curve. The spread does not show a clear increase as fitted level increases. The largest absolute residual is approximately **0.03516101 m**, which exceeds the logger resolution of 0.01 m by about 0.02516 m. Therefore, the fit is strong overall but does not reproduce every short-term variation in the raw logger record.

5. Area Under the Fitted Curve

Quantity	Result
Fitted-curve area from scipy.integrate.quad	341.04332844 m·h
quad absolute error estimate	$3.7863\times10^{\blacksquare 12}$ m·h
Raw trapezoidal cross-check	341.04750000 m·h
Absolute difference	0.00417156 m·h

Area interpretation: The integral has units of m·h and represents accumulated reservoir stage over the one-day observation period. For a flood-control office, it can be used as a compact measure of the reservoir's overall stage exposure during the period; it is not a volume unless an appropriate reservoir area relationship is also supplied.

The two area values differ slightly because quad integrates the smooth fitted logistic function, whereas the trapezoidal calculation connects the discrete raw observations with straight-line segments.

6. Summary

The four-parameter logistic model gives $R^2 = 0.97744068$ with a standard error of estimate of 0.01421877 m. The maximum continuous fitted filling rate is 0.01455752 m/h at 10.99008 h. The fitted-curve area over 0 to 23.75 h is 341.04332844 m·h. The residual diagnostics indicate a generally strong fit with some systematic departures from the raw logger record.